Mohammad Rajabi

Website | Github | LinkedIn | Email | Amsterdam, Netherlands

Summary

AI Engineer with 2 years of experience building and shipping **LLM-powered** solutions — production RAG pipelines, **agentic AI** systems with LangGraph, vector retrieval, and NLP analytics. Master's in AI with thesis on LLM-driven tool orchestration. Holder of a Dutch Orientation Year (Zoekjaar) visa, available from June 2026.

Work Experience

AI Engineer — *July 2025 – Present*

Pet 24-7 Global Ltd — *London, United Kingdom*

Live app: App Store – Google Play

Agentic RAG Platform

- Built the chatbot as a **LangGraph ReAct agent** with pluggable tools for knowledge base retrieval, semantic history search, multimodal OCR, image generation, web search, and PDF generation — enabling autonomous multi-step veterinary consultations.
- Engineered a **hybrid memory architecture** combining a PostgreSQL session cache with Milvus vector search for both fast recent-turn recall and long-term semantic history retrieval.
- Integrated **Langfuse** for end-to-end LLM observability across all agent steps and tool calls.
- Built **MCP servers** to expose agent tools as modular, standardized interfaces.

Production RAG System

- Built and deployed a **production RAG veterinary chatbot** with role-based logic for pet owners and veterinarians, integrating semantic retrieval, cross-encoder re-ranking, and multilingual query handling over a domain-specific knowledge base.
- Designed a **per-user episodic memory system** using vector embeddings to store and semantically retrieve past clinical cases, enabling context-aware follow-up responses across sessions.
- Delivered **real-time SSE streaming** and a multimodal pipeline covering OCR, speech-to-text, and PDF generation.

RAG Retrieval Benchmark

- Benchmarked **Milvus, Qdrant, and Weaviate** across chunking strategies (recursive, semantic, fixed-size), evaluating latency, throughput, and retrieval quality.

AI Analytics Dashboard

- Built an end-to-end **NLP analytics pipeline** processing all chatbot conversations through a two-layer NER system (domain dictionary + biomedical token classifier) and BERTopic-based topic modeling with LLM-generated labels.
- Developed an **AI-powered analytics dashboard** providing real-time NLP insights — entity trends, topic modeling, co-occurrence analysis, and regional breakdowns — via a React frontend and FastAPI backend.

Full Stack Developer — *Nov 2023 – Nov 2024*

MeNEW - Personal Digital Menu — *Part-time*

Digital Menu Platform

- Built and launched a full-stack digital menu platform using Next.js (TypeScript), Django REST API, and GraphQL, with CI/CD pipelines for automated testing and deployment.

Technical Skills

Languages	Python, Java, C++, JavaScript, TypeScript
AI/ML	NLP, LLM, RAG, Agentic AI, Deep Learning, Computer Vision, Classification, Regression, Clustering
Frameworks	PyTorch, TensorFlow, Hugging Face, LangChain, LangGraph, FastAPI, Keras, Spark
Libraries	Pandas, NumPy, NLTK, SciPy, BERTopic, Matplotlib, Plotly, Seaborn
Vector Databases	Milvus, Qdrant, Weaviate, ChromaDB
Cloud & Storage	AWS S3, Google Cloud (Vertex AI)
LLM APIs	Google Gemini, OpenAI API, MCP
Web & DevOps	React, Next.js, Node.js, Django, Docker, Git, SQL, MongoDB, PostgreSQL, Prisma, JWT

Education

University of Padua | **Padova, Italy**
*Master in Computer Science (Major in **Artificial Intelligence**)* | *Sep 2022 – Sep 2025*
GPA: **105/110**

Thesis: Improving Navigation in LLM-Based AI Frameworks *October 2024 – July 2025*

Built an **agentic framework** where an LLM serves as the reasoning engine, orchestrating domain-specific tools via structured APIs and **in-context prompting**. The LLM generates executable code at runtime to select and compose tools, observes intermediate results, and iterates toward a solution — following a **ReAct-style reasoning loop**. Designed the prompt architecture, extended the tool suite, and benchmarked the system against end-to-end and modular baselines.

Supervisor: Prof. Lamberto Ballan

Course Projects

- **Sentiment Analysis and Topic Modeling with Fine-tuned Transformer Models** *(Code and report on GitHub)*
- **Image Colorization with Conditional GANs and Pretrained U-Net** *(Code and report on GitHub)*
- **CNN and Vision Transformer Fine-tuning for Object Detection** *(Code and report on GitHub)*
- **LLM Pruning and Fine-Tuning for Efficient Text Summarization**

For other works, explore more on my *GitHub*

Languages

English: C2 **Dutch:** A2 **Italian:** A2 **Persian:** Native